Network Traffic Analysis Report (Wireshark)

1. Objective

To capture and analyze network traffic using Wireshark, identify HTTP/DNS activity, and detect unusual or malformed packets for security assessment.

2. Methodology

```
Filters Used:
http.request.method == "GET"
malformed
tcp.analysis.flags
tcp.analysis.retransmission
tcp.flags.reset == 1
```

3. Key Findings

1. DNS queries and responses observed for chrome.cloudflare-dns.com
2. TCP SYN packets indicating connection attempts to external servers
3. TCP RST packets indicating abrupt connection resets
4. TCP retransmissions suggesting network instability or packet loss
5. QUIC protocol traffic observed (encrypted HTTP/3 traffic)
6. SSDP multicast traffic observed (local network discovery)
7. HTTP GET requests not visible due to HTTPS encryption

4. Security Analysis

Observation	Interpretation
DNS Traffic	Normal domain resolution
TCP SYN	Connection initiation / scanning
TCP RST	Connection reset
Retransmissions	Network instability
QUIC	Encrypted traffic (not inspectable)

5. Evidence (Screenshots)

Figure 1

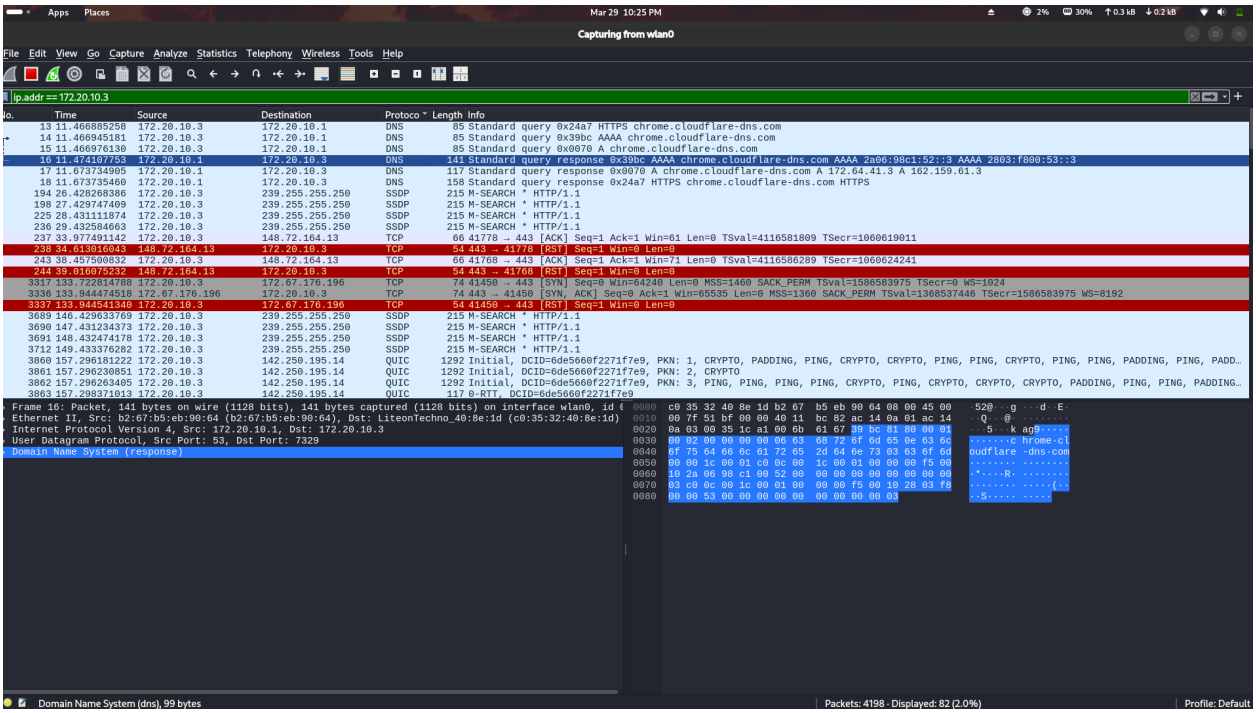


Figure 2

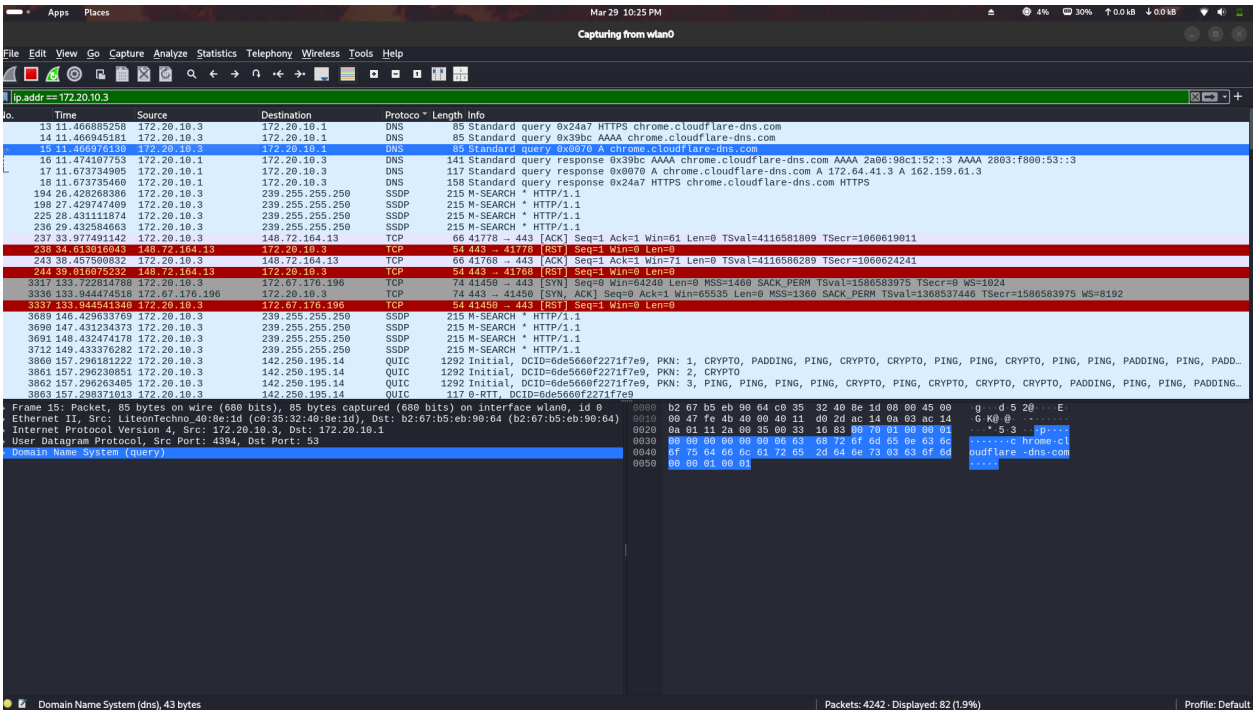
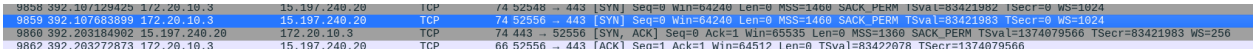


Figure 3



6. Conclusion

The analysis identified multiple network behaviors including DNS resolution, TCP handshakes, retransmissions, and encrypted QUIC traffic. Observed anomalies such as RST packets and retransmissions indicate potential network issues or scanning activity. HTTPS encryption prevented direct inspection of HTTP GET requests.